An Overview of Randomized Singular Value Decomposition (rSVD) Applications

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May 22, 2026

Abstract

Low-rank matrix approximation techniques play a central role in scientific computing, data analysis, and reduced-order modeling. Among these techniques, the Singular Value Decomposition (SVD) provides optimal low-rank approximations but may become computationally expensive for large-scale problems. To address these limitations, randomized algorithms such as Randomized Singular Value Decomposition (rSVD) have emerged as efficient alternatives capable of approximating dominant matrix subspaces with substantially reduced computational cost.

This work presents an overview of randomized Singular Value Decomposition methods and their applications. The mathematical foundations of classical SVD, randomized SVD, oversampling strategies, and power iteration techniques are discussed in detail. The theoretical analysis is complemented by numerical experiments involving image compression and reduced-order modeling for nonlinear partial differential equations.

The practical section investigates the performance of deterministic POD-SVD, randomized POD-rSVD, POD-DEIM, and POD-rSVD-DEIM methods applied to the Allen–Cahn equation and to a nonlinear steady-state elliptic benchmark problem. Particular attention is devoted to the influence of the oversampling parameter and the number of power iterations on approximation quality and computational efficiency.

The numerical experiments show that randomized SVD can reproduce deterministic low-rank approximations with high accuracy while substantially reducing the computational cost associated with basis construction. The results also demonstrate that randomized techniques can be naturally integrated with nonlinear hyper-reduction methods such as DEIM, providing an effective framework for scalable reduced-order modeling.

1 Introduction and Definitions

Low-rank matrix approximations play a fundamental role in modern scientific computing, data analysis, machine learning, and signal processing. Many high-dimensional datasets exhibit an intrinsic low-dimensional structure, meaning that most of the relevant information can be represented using only a small number of dominant components. Among the available matrix factorization techniques, the Singular Value Decomposition (SVD) represents one of the most important and widely used tools for extracting these dominant structures.

The SVD provides an optimal low-rank approximation of a matrix in both the spectral and Frobenius norms. For this reason, it is extensively employed in dimensionality reduction, image compression, principal component analysis, reduced-order modeling, and numerical simulations of partial differential equations. However, the computation of the exact SVD becomes computationally expensive for large-scale matrices arising in modern applications.

To address these limitations, randomized algorithms have been developed to efficiently approximate low-rank decompositions. Among these methods, Randomized Singular Value Decomposition (rSVD) has emerged as one of the most successful approaches due to its favorable balance between computational efficiency and approximation accuracy [3]. The central idea of randomized methods is to exploit random projections in order to identify a low-dimensional subspace capturing most of the dominant action of the matrix, thereby reducing the computational cost associated with classical deterministic algorithms.

This work focuses on the theoretical foundations and practical applications of randomized SVD techniques. Particular attention is devoted to the role of oversampling parameters and their influence on approximation errors. The analysis is supported by numerical experiments involving image compression and reduced-order modeling techniques such as Proper Orthogonal Decomposition (POD) and POD-DEIM for partial differential equations.

The remainder of this chapter introduces the mathematical foundations of classical Singular Value Decomposition, randomized SVD methods, and the role of oversampling in probabilistic low-rank approximations.

1.1 Singular Value Decomposition

Let $A \in \mathbb{R}^{m\times n}$ be a real matrix with rank $r = \mathrm{rank}(A)$. The Singular Value Decomposition (SVD) of A is a factorization of the form

$$A = U\Sigma V^T, \tag{1}$$

where:

- $U \in \mathbb{R}^{m\times m}$ is an orthogonal matrix whose columns are called the left singular vectors of A,
- $V \in \mathbb{R}^{n\times n}$ is an orthogonal matrix whose columns are called the right singular vectors of A,
- $\Sigma \in \mathbb{R}^{m\times n}$ is a diagonal matrix containing the singular values of A.

The diagonal entries of Σ are non-negative real numbers

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0, \tag{2}$$

called the singular values of A, while all remaining diagonal entries are zero. The singular values satisfy

$$\sigma_i = \sqrt{\lambda_i(A^T A)}, \tag{3}$$

where $\lambda_i(A^T A)$ denotes the eigenvalues of the symmetric positive semidefinite matrix $A^T A$ ordered in decreasing magnitude.

The SVD exists for every real or complex matrix and provides an orthogonal decomposition of the associated linear transformation. In particular, the columns of U form an orthonormal basis for the column space of A, while the columns of V form an orthonormal basis for the row space of A [2, 6].

A particularly important application of the SVD is the construction of low-rank approximations. The truncated rank-k SVD of A is defined as

$$A_k = U_k \Sigma_k V_k^T, \tag{4}$$

where U_k, Σ_k, and V_k contain only the first k singular vectors and singular values. The Eckart–Young–Mirsky theorem states that A_k is the optimal rank-k approximation of A with respect to both the spectral norm and the Frobenius norm:

$$A_k = \arg \min_{\mathrm{rank}(B)\leq k} \|A - B\|. \tag{5}$$

Moreover,

$$\|A - A_k\|_2 = \sigma_{k+1}, \tag{6}$$

and

$$\|A - A_k\|_F^2 = \sum_{j=k+1}^{r} \sigma_j^2. \tag{7}$$

Due to these optimality properties, the SVD plays a central role in dimensionality reduction, principal component analysis, signal processing, scientific computing, and data compression. However, for large-scale matrices, the computation of the exact SVD may become computationally prohibitive, motivating the development of approximate and randomized techniques.

1.2 Randomized SVD

Randomized Singular Value Decomposition (Randomized SVD or rSVD) is a probabilistic algorithm designed to efficiently compute low-rank approximations of large matrices [3]. The main idea is to identify a low-dimensional subspace that captures most of the action of the input matrix using random projections, and then compute the decomposition in this reduced subspace.

Let $A \in \mathbb{R}^{m\times n}$ and let $k \ll \min(m, n)$ denote the target rank. Instead of directly computing the exact truncated SVD of A, randomized methods first construct an approximate orthonormal basis for the range of A.

A random test matrix

$$\Omega \in \mathbb{R}^{n\times(k+p)} \tag{8}$$

is generated, where p is a small oversampling parameter introduced to improve approximation quality. Typically, the entries of Ω are independently sampled from a Gaussian distribution.

The matrix

$$Y = A\Omega \tag{9}$$

is then computed. The columns of Y span, with high probability, the dominant column space of A. An orthonormal basis Q for the range of Y is obtained through a QR decomposition:

$$Y = QR. \tag{10}$$

The original matrix is subsequently projected onto the lower-dimensional subspace:

$$B = Q^T A, \tag{11}$$

where $B \in \mathbb{R}^{(k+p)\times n}$ is significantly smaller than A. A deterministic SVD is then applied to B:

$$B = \widetilde{U}\Sigma V^T. \tag{12}$$

Finally, an approximate factorization of A is recovered as

$$A \approx Q\widetilde{U}\Sigma V^T. \tag{13}$$

The computational advantage of randomized SVD arises from the fact that the expensive decomposition is performed on the compressed matrix B rather than directly on A. For dense matrices, randomized algorithms can achieve computational complexity approximately proportional to

$$\mathcal{O}(mn \log k), \tag{14}$$

compared to the classical complexity

$$\mathcal{O}(mnk) \tag{15}$$

required by traditional deterministic methods [3].

To improve approximation accuracy in cases where the singular values decay slowly, power iteration schemes can be employed. The sampled matrix becomes

$$Y = (AA^T)^q A\Omega, \tag{16}$$

where $q \geq 0$ denotes the number of power iterations. This procedure amplifies the contribution of dominant singular directions and improves the quality of the approximation.

Randomized SVD offers several important advantages:

- reduced computational cost for large-scale matrices,
- lower memory requirements,
- improved scalability on parallel and distributed architectures,
- efficient processing of sparse or streaming data,
- reduced number of passes over the data matrix.

These properties make randomized techniques particularly suitable for modern large-scale applications in machine learning, scientific computing, and data analysis.

Nevertheless, randomized SVD also presents several limitations. Since the algorithm is probabilistic, the resulting decomposition is approximate rather than exact. The approximation quality depends on the singular value decay, the oversampling parameter, and the number of power iterations. Furthermore, randomized methods may become less advantageous for small matrices or when extremely high numerical precision is required.

Despite these limitations, randomized SVD has become one of the most widely adopted approaches for scalable low-rank matrix approximation due to its strong balance between computational efficiency and approximation accuracy [3].

1.3 Oversampling and Approximation Error

A key component of randomized SVD algorithms is the use of oversampling. Instead of constructing a subspace of dimension exactly equal to the target rank k, randomized methods typically employ a slightly larger subspace of dimension $k + p$, where $p \geq 0$ is called the oversampling parameter.

The purpose of oversampling is to improve the probability that the randomly generated subspace accurately captures the dominant singular directions of the matrix. In practical applications, small oversampling values such as $p = 5$ or $p = 10$ are often sufficient to significantly improve approximation quality [3].

Given a matrix $A \in \mathbb{R}^{m\times n}$, randomized SVD constructs an approximation

$$A \approx QQ^T A, \tag{17}$$

where the columns of Q form an orthonormal basis for the sampled subspace. The approximation error is therefore commonly measured using either the spectral norm

$$\|A - QQ^T A\|_2, \tag{18}$$

or the Frobenius norm

$$\|A - QQ^T A\|_F. \tag{19}$$

Theoretical analysis shows that the expected approximation error depends on the decay of the singular values and on the oversampling parameter. In particular, increasing p improves the probability that the dominant singular subspace is accurately approximated.

For Gaussian random matrices, Halko et al. [3] derived probabilistic bounds showing that, with high probability, the approximation error remains close to the optimal truncated SVD error. A simplified version of the expected spectral norm bound can be expressed as

$$\mathbb{E}\left[\|A - QQ^T A\|_2\right] \leq \left[1 + 4\sqrt{\frac{k+p}{p-1}}\right] \sigma_{k+1}, \tag{20}$$

where σ_{k+1} denotes the first neglected singular value.

These results demonstrate that randomized SVD can achieve near-optimal low-rank approximations while substantially reducing computational cost. Nevertheless, the approximation quality may deteriorate when the singular values decay slowly or when insufficient oversampling is employed.

In addition to oversampling, power iteration techniques can further reduce approximation errors by amplifying dominant singular directions. However, increasing the number of power iterations also increases the computational complexity and may introduce additional numerical stability considerations.

The trade-off between computational efficiency, oversampling size, and approximation accuracy constitutes one of the central aspects of randomized low-rank approximation methods and will be analyzed experimentally in the following chapters.

2 Algorithms and Implementation

This section presents the algorithmic foundations of classical truncated Singular Value Decomposition and Randomized Singular Value Decomposition. Particular attention is devoted to the computational structure of the algorithms, their complexity, and the role of oversampling and power iterations in randomized methods.

2.1 Classical Truncated SVD

Given a matrix $A \in \mathbb{R}^{m \times n}$, the truncated Singular Value Decomposition computes the best rank-k approximation of A:

$$A_k = U_k \Sigma_k V_k^T, \tag{21}$$

where:

- $U_k \in \mathbb{R}^{m \times k}$ contains the first k left singular vectors,
- $\Sigma_k \in \mathbb{R}^{k \times k}$ contains the k largest singular values,
- $V_k \in \mathbb{R}^{n \times k}$ contains the first k right singular vectors.

The truncated SVD minimizes the approximation error among all rank-k matrices:

$$A_k = \arg \min_{\mathrm{rank}(B) \leq k} \|A - B\|. \tag{22}$$

In practical implementations, the truncated SVD is commonly computed using bidiagonalization methods followed by iterative eigensolvers or Krylov subspace techniques [2].

For dense matrices, the computational complexity of classical methods is approximately

$$\mathcal{O}(mnk), \tag{23}$$

which may become prohibitively expensive for large-scale applications.

The pseudocode of the truncated SVD procedure is summarized in Algorithm 1.

Algorithm 1 Classical Truncated SVD

Require: Matrix $A \in \mathbb{R}^{m\times n}$, target rank k
Ensure: Rank-k approximation A_k
1: Compute the Singular Value Decomposition: $A = U\Sigma V^T$
2: Retain only the first k singular values and vectors: U_k, Σ_k, V_k
3: Construct the truncated approximation: $A_k = U_k \Sigma_k V_k^T$
4: **return** A_k

2.2 Randomized SVD Algorithm

Randomized Singular Value Decomposition reduces the computational cost of low-rank approximation by projecting the original matrix onto a randomly generated low-dimensional subspace [3].

Let $A \in \mathbb{R}^{m\times n}$ and let k denote the target rank. Instead of directly computing the SVD of A, randomized methods first approximate the range of A using random sampling.

A random matrix $\Omega \in \mathbb{R}^{n\times(k+p)}$ is generated, where p is the oversampling parameter.

The sampled matrix $Y = A\Omega$ captures the dominant action of A with high probability.

An orthonormal basis Q for the range of Y is computed using QR decomposition: $Y = QR$.

The original matrix is then projected onto the reduced subspace: $B = Q^T A$.

Since B has significantly smaller dimensions than A, its SVD can be computed efficiently: $B = \widetilde{U}\Sigma V^T$.

Finally, the approximate singular vectors of A are reconstructed as $U \approx Q\widetilde{U}$,
leading to the approximate factorization

$$A \approx Q\widetilde{U}\Sigma V^T. \tag{24}$$

The complete randomized SVD procedure is described in Algorithm 2.

Algorithm 2 Randomized Singular Value Decomposition

Require: Matrix $A \in \mathbb{R}^{m\times n}$, target rank k, oversampling parameter p
Ensure: Approximate rank-k decomposition of A
1: Generate a Gaussian random matrix: $\Omega \in \mathbb{R}^{n\times(k+p)}$
2: Form the sampled matrix: $Y = A\Omega$
3: Compute the QR decomposition: $Y = QR$
4: Project the matrix onto the reduced subspace: $B = Q^T A$
5: Compute the SVD of the reduced matrix: $B = \widetilde{U}\Sigma V^T$
6: Recover approximate left singular vectors: $U = Q\widetilde{U}$
7: **return** U, Σ, V

2.3 Oversampling and Power Iterations

The oversampling parameter p plays a fundamental role in randomized SVD algorithms. Rather than constructing a subspace of dimension exactly equal to the target rank k, a slightly larger subspace of dimension $k + p$ is employed in order to improve approximation accuracy.

Oversampling reduces the probability that relevant singular directions are missed during random sampling. In practical applications, values such as $p = 5$ or $p = 10$ are typically sufficient to obtain high-quality approximations [3].

When the singular values decay slowly, the approximation quality of the randomized method may deteriorate. To address this issue, power iteration techniques are commonly employed.

The randomized sampling step becomes

$$Y = (AA^T)^q A\Omega, \tag{25}$$

where $q \geq 0$ denotes the number of power iterations.

This procedure amplifies dominant singular directions and attenuates smaller singular components, improving the accuracy of the approximation. However, increasing the number of power iterations also increases computational cost and may introduce additional numerical stability issues.

The approximation error of randomized SVD depends on:

- the decay rate of the singular values,
- the target rank k,
- the oversampling parameter p,
- the number of power iterations q.

Theoretical results show that randomized SVD can achieve near-optimal low-rank approximations with high probability while substantially reducing computational complexity [3].

2.4 Computational Complexity

One of the principal advantages of randomized SVD lies in its computational efficiency.

For a dense matrix $A \in \mathbb{R}^{m \times n}$, the classical truncated SVD typically requires computational complexity of order

$$\mathcal{O}(mnk). \tag{26}$$

In contrast, randomized SVD approximately requires

$$\mathcal{O}(mn \log k), \tag{27}$$

depending on the implementation details and the number of power iterations [3].

Moreover, randomized methods are particularly advantageous in scenarios involving:

- sparse matrices,
- streaming data,
- distributed architectures,
- out-of-core computations,
- large-scale scientific simulations.

Another important advantage is that randomized methods typically require only a small number of passes over the input matrix, making them well suited for modern high-dimensional datasets where memory access represents a major computational bottleneck.

3 Image Compression and Reconstruction

Image compression represents one of the classical applications of low-rank matrix approximation techniques. Digital images can be interpreted as matrices whose entries correspond to pixel intensity values. The Singular Value Decomposition provides an efficient mechanism for approximating these matrices using a reduced number of dominant singular components, enabling compression while preserving the principal visual structures of the image.

This section presents a comparison between classical truncated SVD and Randomized Singular Value Decomposition for image reconstruction. Particular attention is devoted to the role of the oversampling parameter and its influence on approximation quality.

3.1 Mathematical Formulation

Consider a grayscale image represented by a matrix

$$A \in \mathbb{R}^{m \times n}, \tag{28}$$

where each entry a_{ij} corresponds to the intensity value of a pixel.

Applying the Singular Value Decomposition yields

$$A = U\Sigma V^T. \tag{29}$$

The rank-k approximation of the image is obtained through the truncated SVD

$$A_k = U_k \Sigma_k V_k^T, \tag{30}$$

where only the first k singular values and singular vectors are retained.

Since the singular values are ordered by decreasing magnitude, the dominant visual information of the image is concentrated in the first singular components. Consequently, low-rank approximations can reconstruct the main image structures while discarding less relevant information and noise.

The compression ratio can be estimated by comparing the number of stored coefficients. The original image requires storage proportional to

$$mn, \tag{31}$$

while the truncated SVD representation requires approximately

$$k(m + n + 1) \tag{32}$$

coefficients.

When $k \ll \min(m, n)$, significant compression can be achieved.

3.2 Randomized Image Reconstruction

In the randomized framework, the low-rank approximation is computed using Randomized Singular Value Decomposition. Instead of directly decomposing the image matrix, the dominant subspace is estimated through random projections.

Given a target rank k and an oversampling parameter p, the randomized algorithm constructs a low-dimensional basis

$$Q \in \mathbb{R}^{m \times (k+p)} \tag{33}$$

that approximates the dominant column space of the image matrix. The compressed matrix

$$B = Q^T A \tag{34}$$

is then factorized using a deterministic SVD, yielding the approximate reconstruction

$$A \approx Q\widetilde{U}\Sigma V^T. \tag{35}$$

The primary advantage of randomized reconstruction is the reduction of computational cost for large images, especially when only a relatively small number of singular components is required.

3.3 Experimental Setup

The numerical experiment was designed to compare classical truncated SVD and Randomized Singular Value Decomposition in the context of grayscale image compression and reconstruction.

A grayscale image was loaded and normalized so that pixel intensities belonged to the interval $[0, 1]$. The image matrix was then approximated using both deterministic truncated SVD and randomized SVD for different target ranks.

The following target ranks were considered:

$$k \in \{5,\ 10,\ 25,\ 50,\ 100\}. \tag{36}$$

These values were selected in order to analyze the progressive reconstruction of image structures as the number of retained singular components increases. Very small values of k generate highly compressed representations with substantial information loss, while larger values preserve increasingly finer image details.

For the randomized SVD experiments, the following oversampling parameters were employed:

$$p \in \{5,\ 10,\ 20\}. \tag{37}$$

The oversampling parameter determines the number of additional random directions sampled beyond the target rank. Small oversampling values are typically sufficient to substantially improve approximation robustness while maintaining low computational complexity [3].

A single power iteration was employed:

$$q = 1. \tag{38}$$

Power iterations improve approximation accuracy by amplifying dominant singular directions, especially in problems where singular values decay slowly.

The randomized approximation therefore computes a subspace of dimension

$$k + p, \tag{39}$$

rather than exactly k dimensions.

The reconstruction quality was evaluated using the relative Frobenius residual:

$$E_{\text{rel}} = \frac{\|A - \widehat{A}\|_F}{\|A\|_F}, \tag{40}$$

where:

- A denotes the original image matrix,
- $\widehat{A}$ denotes the reconstructed approximation,
- $\|\cdot\|_F$ denotes the Frobenius norm.

The Frobenius residual measures the global reconstruction error over all pixels and provides a quantitative estimate of approximation quality.

In addition to reconstruction error, the computational execution time was measured for each decomposition. Execution times were evaluated using wall-clock timing functions immediately before and after the decomposition routines. The reported values therefore include the complete computation of the low-rank approximation.

The deterministic SVD implementation employed the standard singular value decomposition routine available in NumPy, while randomized SVD was computed using the `randomized_svd` implementation available in Scikit-Learn.

3.4 Numerical Results

Figure 1 presents the comparison between classical truncated SVD and randomized SVD reconstructions.

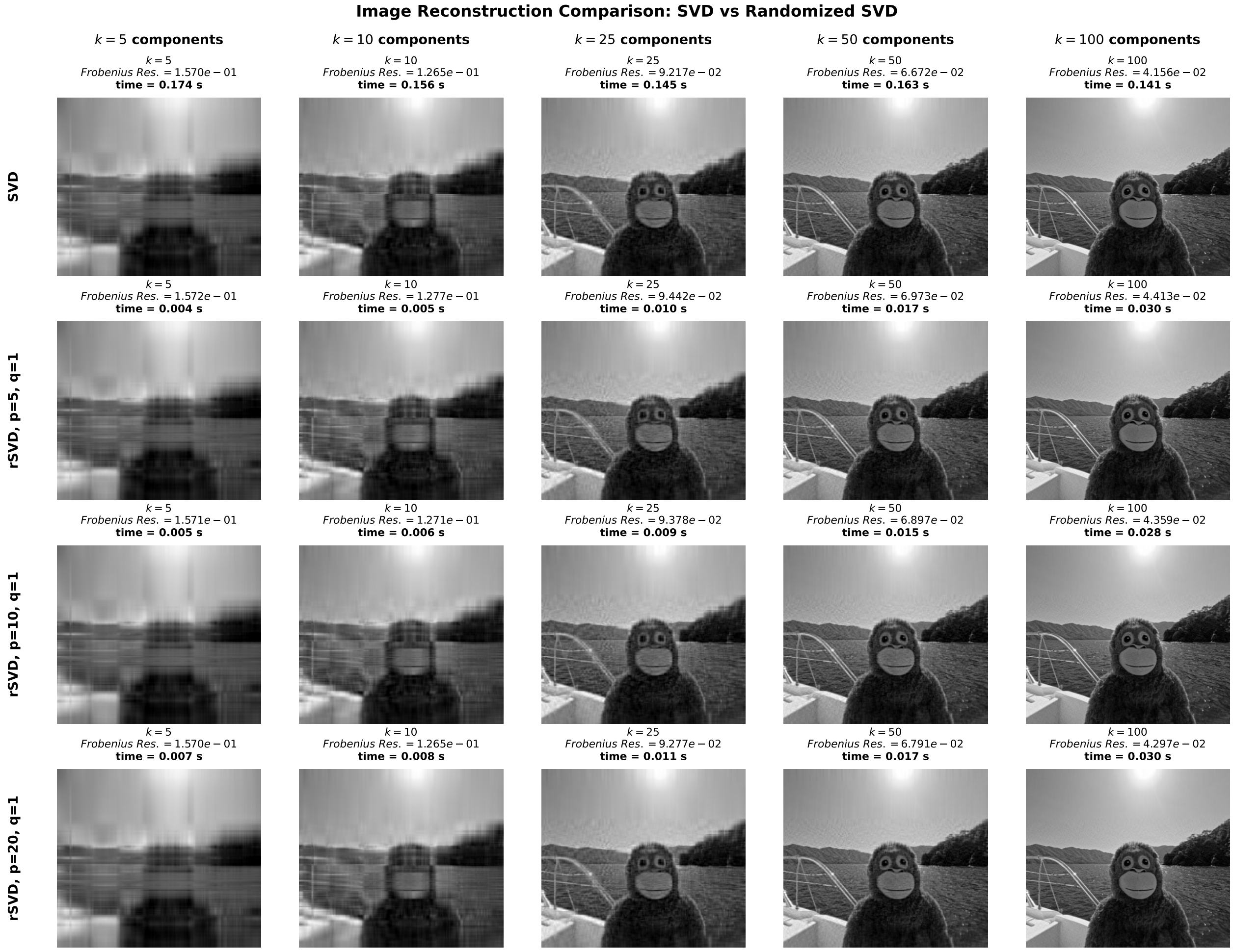

Figure 1: Comparison between deterministic truncated SVD and randomized SVD image reconstructions for different target ranks k and oversampling parameters p. The reported values correspond to the relative Frobenius residual and computational execution time.

The first row corresponds to deterministic truncated SVD, while the remaining rows correspond to randomized SVD reconstructions with increasing oversampling parameters.

The columns correspond to increasing target ranks:

$$k \in \{5,\ 10,\ 25,\ 50,\ 100\}. \tag{41}$$

As expected, increasing the target rank substantially improves reconstruction quality for both deterministic and randomized methods. For small values of k, only the dominant low-frequency image structures are preserved, while fine details are progressively recovered as additional singular components are included.

The randomized SVD reconstructions exhibit approximation errors comparable to those obtained with deterministic SVD. In particular, increasing the oversampling parameter improves the stability and accuracy of the randomized approximation, reducing the probability of missing dominant singular directions.

The results also demonstrate the computational advantage of randomized methods. For all tested target ranks, randomized SVD achieved significantly lower execution times than deterministic SVD while preserving comparable reconstruction quality.

The relative Frobenius residual decreases monotonically as the target rank increases. This behavior is consistent with the theoretical optimality properties of low-rank approximations, since additional singular components capture increasingly finer image information.

The influence of oversampling becomes more evident at intermediate approximation ranks. Larger oversampling values generally produce slightly lower reconstruction errors at the expense of increased computational cost due to the larger sampled subspace dimension.

Overall, the experiment confirms that randomized SVD provides an effective trade-off between computational efficiency and reconstruction accuracy. These results are consistent with the theoretical properties discussed in [3], demonstrating that randomized low-rank approximation methods can achieve near-optimal reconstruction quality while substantially reducing computational complexity.

4 PDE Model Reconstruction

Reduced-order modeling techniques play a fundamental role in the numerical approximation of large-scale dynamical systems arising from partial differential equations (PDEs). High-fidelity discretizations of nonlinear PDEs often generate systems with extremely large dimensions, leading to substantial computational costs in simulation, optimization, parameter estimation, and uncertainty quantification problems.

Proper Orthogonal Decomposition (POD) provides an efficient framework for constructing low-dimensional reduced spaces capable of approximating the dominant dynamics of the full-order system. However, for nonlinear problems, the computational complexity associated with nonlinear terms may still depend on the dimension of the original discretized system. To address this limitation, the Discrete Empirical Interpolation Method (DEIM) is commonly employed to efficiently approximate nonlinear operators [1].

This section presents two model reduction experiments involving nonlinear PDEs. The first problem concerns the Allen–Cahn equation, while the second problem considers a nonlinear steady-state elliptic equation commonly used as a benchmark problem for POD-DEIM methods. The objective of the experiments is to compare deterministic POD-SVD, randomized POD-rSVD, POD-DEIM, and POD-rSVD-DEIM in terms of approximation quality and computational efficiency.

In particular, the analysis focuses on the influence of:

- the reduced basis dimension r,
- the DEIM interpolation dimension m,
- the oversampling parameter p,
- the number of randomized power iterations q.

The randomized methods are based on the randomized SVD framework introduced in [3].

4.1 Proper Orthogonal Decomposition

Proper Orthogonal Decomposition constructs a reduced basis from a collection of solution snapshots generated by the full-order numerical model [5].

Let

$$u_1, u_2, \ldots, u_{N_s} \in \mathbb{R}^N \tag{42}$$

denote a set of snapshots obtained from numerical simulations at different time instances or parameter values. The snapshot matrix is defined as

$$S = \begin{bmatrix} u_1 & u_2 & \cdots & u_{N_s} \end{bmatrix} \in \mathbb{R}^{N \times N_s}. \tag{43}$$

Applying the Singular Value Decomposition to the snapshot matrix yields

$$S = U\Sigma V^T. \tag{44}$$

The POD basis is constructed by retaining the first r dominant left singular vectors:

$$\Phi_r = \begin{bmatrix} u_1 & u_2 & \cdots & u_r \end{bmatrix}. \tag{45}$$

The reduced approximation of the solution is then expressed as

$$u(x,t) \approx \Phi_r a(t), \tag{46}$$

where $a(t) \in \mathbb{R}^r$ contains the reduced coefficients.

The POD basis minimizes the projection error in the L^2 sense and provides the optimal linear reduced representation of the snapshot set.

4.2 Randomized POD

In the randomized setting, the POD basis is constructed using randomized Singular Value Decomposition. Given a snapshot matrix $S \in \mathbb{R}^{N \times N_s}$, the randomized algorithm first generates a Gaussian random matrix

$$\Omega \in \mathbb{R}^{N_s \times (r+p)}, \tag{47}$$

where p is the oversampling parameter. The randomized range finder computes

$$Y = S\Omega, \tag{48}$$

followed by a QR factorization

$$Y = QR. \tag{49}$$

The compressed matrix

$$B = Q^T S \tag{50}$$

is then factorized through a deterministic SVD:

$$B = \widetilde{U}\Sigma V^T. \tag{51}$$

The approximate POD basis becomes

$$U \approx Q\widetilde{U}. \tag{52}$$

To improve the approximation of slowly decaying singular spectra, randomized power iterations may be introduced:

$$Y = (SS^T)^q S\Omega, \tag{53}$$

where $q \geq 0$ denotes the number of power iterations.

The experiments consider:

$$p \in \{0, 5, 20, 50\}, \qquad q \in \{0, 1, 2\}. \tag{54}$$

The influence of these parameters is analyzed through both projection and DEIM approximation errors.

4.3 Discrete Empirical Interpolation Method

Although POD substantially reduces the dimensionality of the system, nonlinear operators may still require evaluations in the full-order space. The Discrete Empirical Interpolation Method (DEIM) addresses this issue by approximating nonlinear terms using interpolation over a reduced set of spatial indices [1].

Let

$$f(u) \in \mathbb{R}^N \tag{55}$$

denote a nonlinear operator. A POD basis for the nonlinear snapshots is first constructed:

$$U_f = \begin{bmatrix} \psi_1 & \psi_2 & \cdots & \psi_m \end{bmatrix}. \tag{56}$$

DEIM approximates the nonlinear function as

$$f(u) \approx U_f c(t), \tag{57}$$

where the coefficients $c(t)$ are determined through interpolation conditions applied at selected spatial indices.

Introducing the selection matrix P, the DEIM approximation becomes

$$f(u) \approx U_f (P^T U_f)^{-1} P^T f(u). \tag{58}$$

The resulting approximation significantly reduces the computational complexity associated with nonlinear evaluations, since only a small subset of entries of the nonlinear function must be computed.

4.4 Allen–Cahn Equation

The first numerical experiment considers the Allen–Cahn equation:

$$y_t(x,t) - \alpha \Delta y(x,t) - \mu \left(y(x,t) - y^3(x,t) \right) = 0, \qquad (x,t) \in \Omega \times (0,T), \tag{59}$$

subject to homogeneous Dirichlet boundary conditions

$$y(a,t) = 0 = y(b,t), \qquad t \in (0,T), \tag{60}$$

and initial condition

$$y(x,0) = y_0(x). \tag{61}$$

The parameters used in the experiments are

$$\alpha = 0.1, \qquad \mu = 11, \qquad T = 2, \qquad \Omega = [0,1] \times [0,1]. \tag{62}$$

The problem is discretized on a uniform two-dimensional grid with

$$n = 256 \tag{63}$$

grid points per spatial direction, corresponding to a full-order dimension of approximately

$$N = 256^2 = 65536. \tag{64}$$

Time integration is performed using a fixed time step

$$\Delta t = 0.005, \tag{65}$$

while snapshots are collected every five time steps.

The Allen–Cahn equation represents a nonlinear reaction-diffusion model commonly used in phase separation and interface dynamics problems. Its nonlinear cubic term makes it an appropriate benchmark for POD-DEIM reduced-order modeling.

The evolution of representative full-order snapshots is shown in Figure 2. The solution rapidly evolves toward a smooth phase-separated configuration dominated by low-frequency spatial structures. This behavior suggests the existence of a strongly compressible low-dimensional manifold, making the problem particularly suitable for POD-based reduced-order modeling.

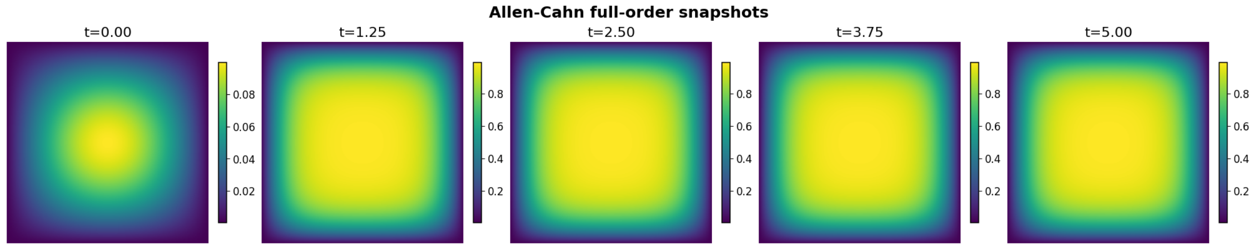

Figure 2: Representative full-order snapshots of the Allen–Cahn equation at different time instances. The solution progressively evolves toward a smooth phase-separated state characterized by dominant low-frequency spatial structures. The strong spatial coherence of the snapshots suggests that the solution manifold is highly compressible and can be accurately represented using a relatively small number of POD modes.

The qualitative structure of the full-order dynamics shown in Figure 2 already suggests a rapidly decaying low-rank structure. This observation is quantitatively confirmed by the POD projection errors reported in Figure 3.

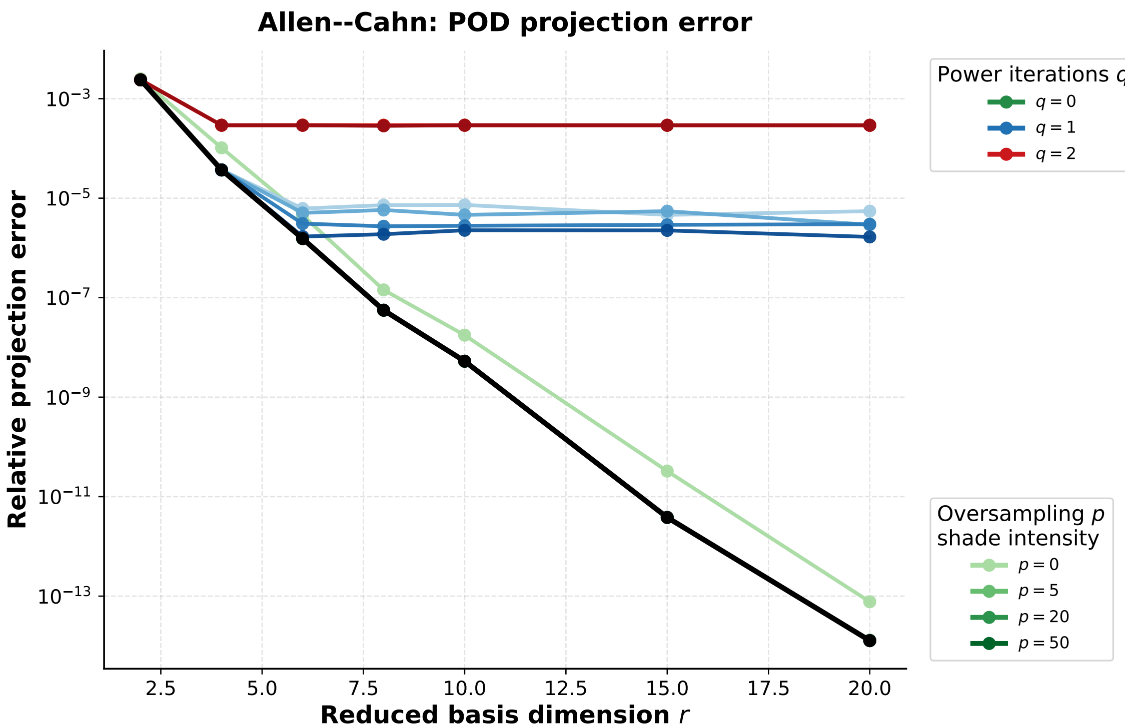

Figure 3: Relative POD projection error for the Allen–Cahn equation. The figure compares deterministic POD-SVD with randomized POD-rSVD for different values of the oversampling parameter p and the number of power iterations q. The randomized approximation with $q = 0$ closely reproduces the deterministic POD error, reaching near machine precision for sufficiently large reduced dimensions.

The results show an extremely rapid decay of the projection error as the reduced basis dimension increases. The deterministic POD-SVD error decreases from approximately 10^{-3} for small reduced dimensions down to values close to machine precision for larger ranks. This behavior indicates that the Allen–Cahn snapshot matrix possesses a strongly low-rank structure. The dominant dynamics of the system are therefore well represented by a relatively small number of POD modes.

The randomized POD-rSVD results exhibit a remarkably similar behavior when $q = 0$. In particular, for moderate and large oversampling values, the randomized approximation reproduces the deterministic POD-SVD projection error almost exactly. This confirms that randomized SVD can efficiently approximate the dominant POD subspace in problems characterized by rapidly decaying singular spectra.

An interesting phenomenon emerges when increasing the number of power iterations. Although randomized power iterations are theoretically intended to improve the approximation quality for slowly decaying singular values, the present experiments show that increasing q actually deteriorates the approximation quality. For $q = 1$, the error stabilizes around 10^{-6}, while for $q = 2$ the approximation deteriorates further, remaining approximately around 10^{-4} even for large reduced dimensions.

This behavior suggests that the Allen–Cahn snapshot matrix already possesses a strongly separated singular spectrum. Additional power iterations therefore over-amplify the dominant singular directions while suppressing smaller singular modes that still contribute to accurate high-rank reconstructions.

The influence of the randomized parameters is further illustrated in the heatmap shown in Figure 4.

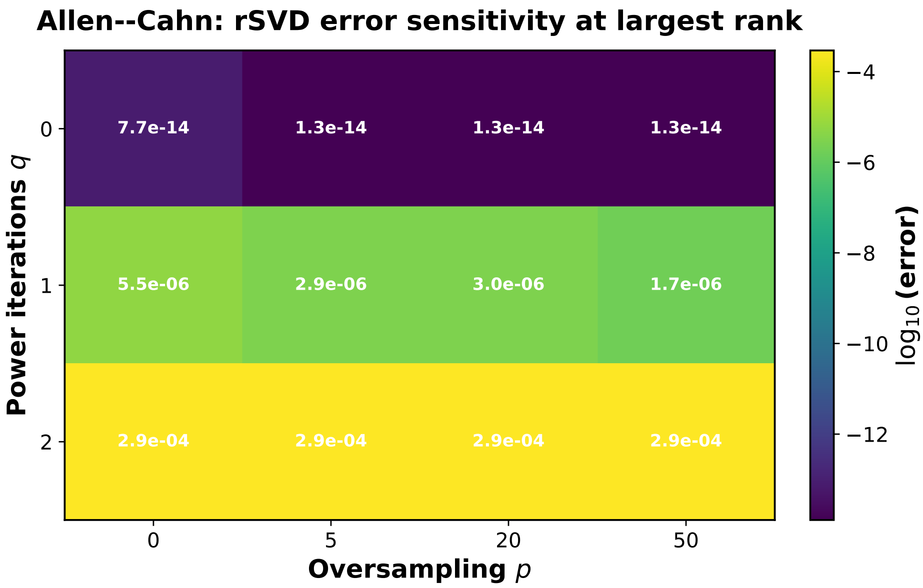

Figure 4: Sensitivity analysis of the randomized POD-rSVD approximation error for the Allen–Cahn equation at the largest reduced dimension considered. The heatmap highlights the influence of the oversampling parameter p and the number of power iterations q. The best approximations are consistently obtained for $q = 0$, while increasing q produces larger reconstruction errors.

The DEIM approximation errors for the Allen–Cahn equation are reported in Figure 5.

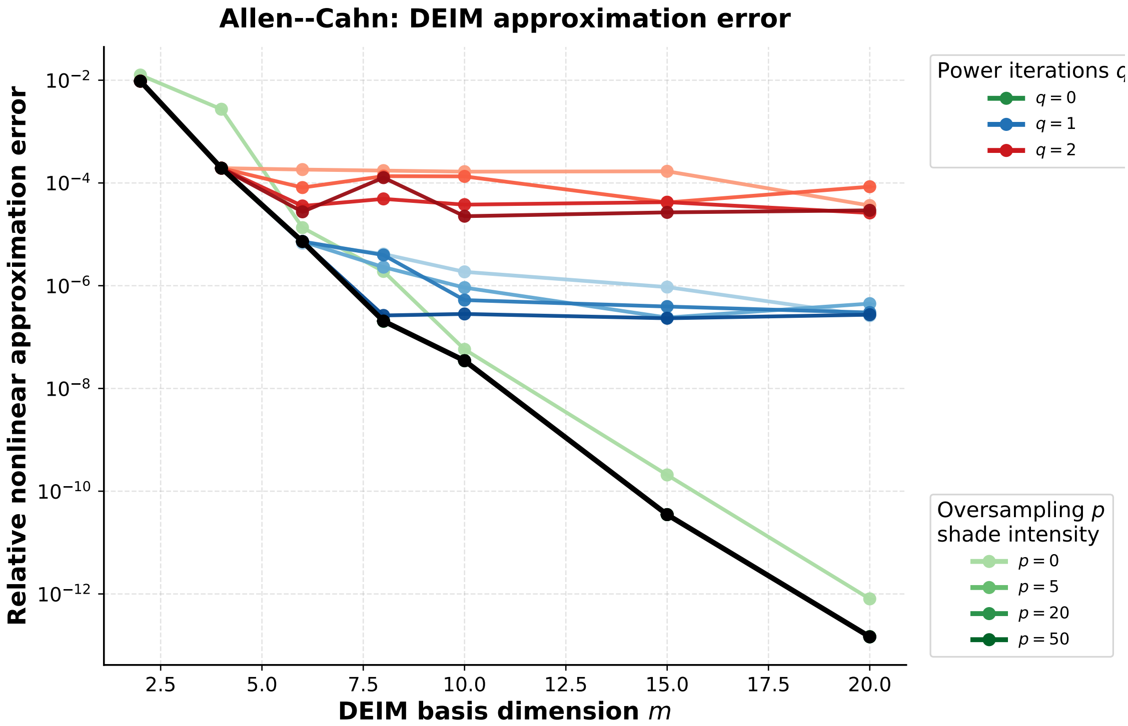

Figure 5: Relative nonlinear approximation error for POD-DEIM and POD-rSVD-DEIM applied to the Allen–Cahn equation. The error rapidly decreases as the DEIM interpolation dimension m increases. The randomized DEIM approximation with $q = 0$ closely reproduces the deterministic POD-DEIM results.

The DEIM approximation error decreases very rapidly as the interpolation dimension m increases. For deterministic POD-DEIM, the nonlinear approximation error reaches values close to machine precision for sufficiently large interpolation dimensions. The randomized POD-rSVD-DEIM approximation exhibits almost identical behavior for $q = 0$, again confirming that randomized low-rank approximations are highly effective for this problem.

The heatmap in Figure 6 confirms the same trend.

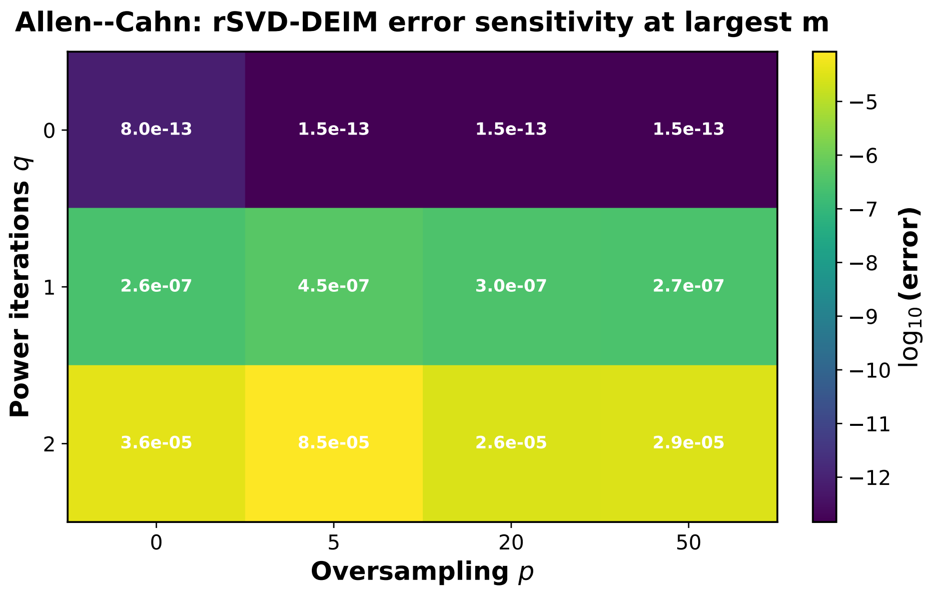

Figure 6: Sensitivity analysis of the POD-rSVD-DEIM nonlinear approximation error for the Allen–Cahn equation. The approximation quality strongly depends on the number of power iterations q, while the influence of the oversampling parameter p is comparatively moderate.

From a computational perspective, randomized POD-rSVD substantially reduces the offline basis construction cost compared to deterministic POD-SVD. The measured execution times indicate speedups ranging approximately from one order of magnitude down to smaller factors depending on the reduced dimension and the selected randomized parameters. However, the exact timings should be interpreted cautiously, since they are affected by hardware characteristics, operating system scheduling, Python overhead, and numerical library implementations.

4.5 Nonlinear Steady-State Elliptic Problem

The second numerical experiment considers the nonlinear steady-state PDE presented in [1]:

$$-\nabla^2 u(x,y) + s(u(x,y);\mu) = 100\sin(2\pi x)\sin(2\pi y), \tag{66}$$

where

$$s(u;\mu) = \frac{\mu_1}{\mu_2}\left(e^{\mu_2 u} - 1\right). \tag{67}$$

The spatial domain is

$$\Omega = (0,1)^2, \tag{68}$$

with parameter vector

$$\mu = (\mu_1, \mu_2) \in [0.01, 10]^2, \tag{69}$$

and homogeneous Dirichlet boundary conditions.

The problem is discretized on a uniform grid with

$$n = 256, \tag{70}$$

again corresponding to a full-order dimension of approximately

$$N = 65536. \tag{71}$$

The nonlinear algebraic system generated by the finite-difference discretization is solved through Newton's method with tolerance

$$10^{-10}. \tag{72}$$

This benchmark problem is particularly suitable for POD-DEIM since the nonlinear exponential term introduces substantial computational complexity in the full-order model.

Representative full-order solutions for different parameter values are shown in Figure 7. Unlike the Allen–Cahn problem, the nonlinear elliptic benchmark exhibits stronger parametric variability and more complex nonlinear spatial structures. This suggests a slower singular value decay and a more challenging reduced-order approximation problem.

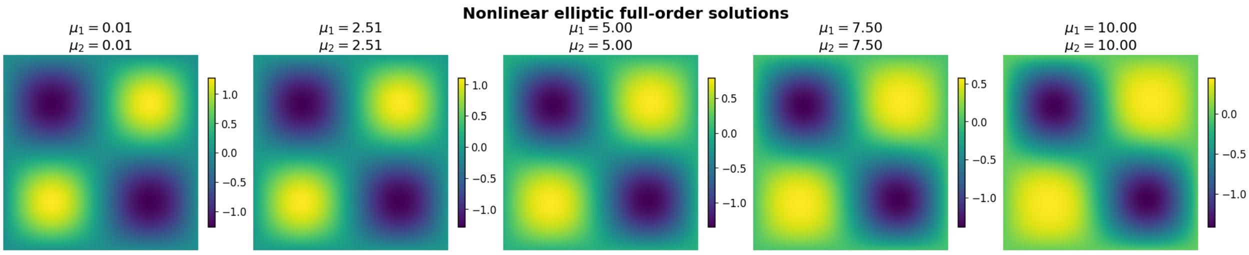

Figure 7: Representative full-order solutions of the nonlinear elliptic benchmark problem for different parameter values (μ_1, μ_2). The exponential nonlinear term generates increasingly complex spatial structures as the parameters vary, producing a richer and less compressible solution manifold compared to the Allen–Cahn equation.

The POD projection errors are shown in Figure 8.

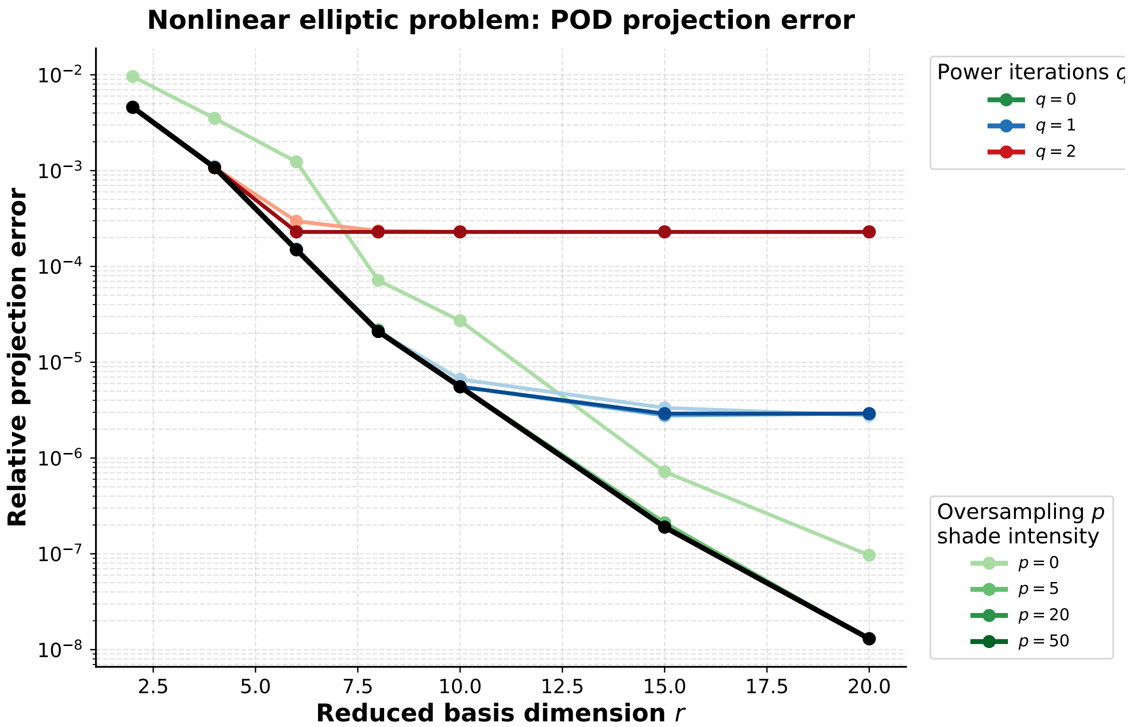

Figure 8: Relative POD projection error for the nonlinear elliptic benchmark problem. Compared to the Allen–Cahn equation, the error decay is slower, indicating a more complex solution manifold.

The stronger variability already visible in the full-order solutions of Figure 7 is reflected in the slower decay of the POD projection error shown in Figure 8. Compared to the Allen–Cahn problem, the elliptic benchmark exhibits a more complex reduced manifold and therefore a more challenging low-rank approximation problem.

The deterministic POD-SVD approximation still converges rapidly, reaching projection errors close to 10^{-8} for sufficiently large reduced dimensions. However, the randomized approximation is noticeably more sensitive to the randomized parameters.

For $q = 0$, randomized POD-rSVD again closely reproduces the deterministic POD-SVD approximation. The influence of the oversampling parameter is more evident in this case. Without oversampling ($p = 0$), the approximation quality deteriorates for small reduced dimensions. Introducing even moderate oversampling significantly improves the reconstruction quality and recovers the deterministic POD-SVD error.

The effect of the randomized parameters is summarized in Figure 9.

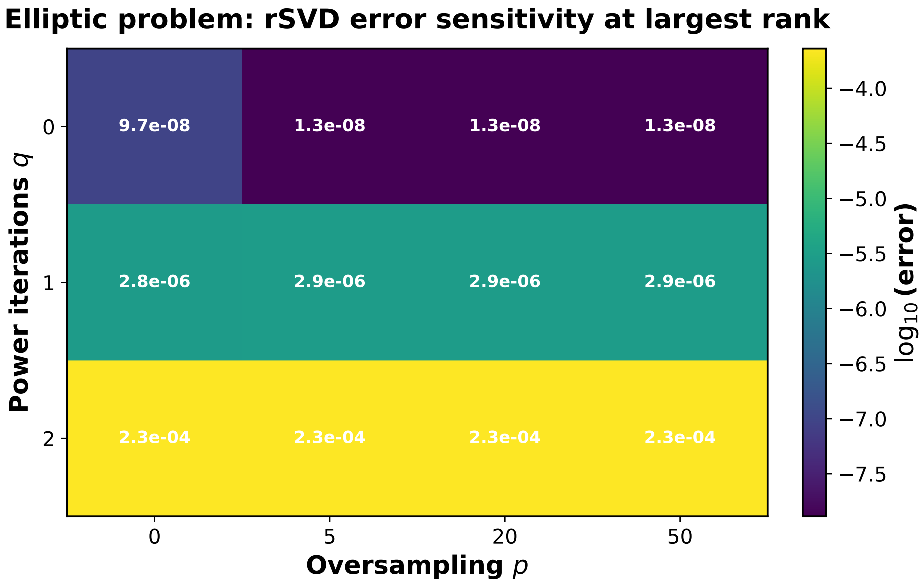

Figure 9: Sensitivity analysis of the randomized POD-rSVD approximation error for the nonlinear elliptic problem. The approximation quality improves significantly when using moderate oversampling, while increasing the number of power iterations again deteriorates the approximation.

Similarly to the Allen–Cahn experiment, increasing the number of power iterations does not improve the approximation. For $q = 1$, the error stabilizes approximately around 10^{-6}, while for $q = 2$ the approximation deteriorates further and remains close to 10^{-4}.

The DEIM approximation errors for the nonlinear elliptic problem are shown in Figure 10.

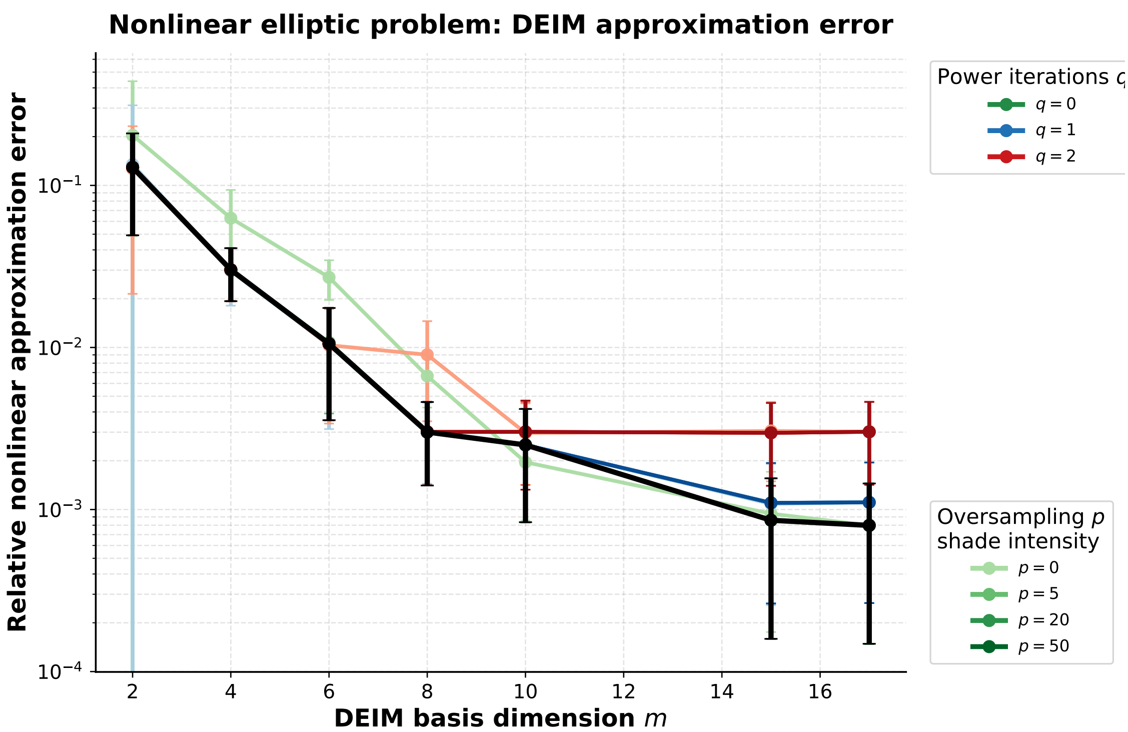

Figure 10: Relative nonlinear approximation error for POD-DEIM and POD-rSVD-DEIM applied to the nonlinear elliptic benchmark problem. The nonlinear approximation converges more slowly than in the Allen–Cahn case due to the exponential nonlinear term.

The DEIM approximation converges more slowly than in the Allen–Cahn experiment. This is expected, since the exponential nonlinear term produces a more complex nonlinear snapshot manifold compared to the cubic Allen–Cahn nonlinearity.

The deterministic POD-DEIM approximation reduces the nonlinear approximation error from approximately 10^{-1} down to around 10^{-3} as the DEIM interpolation dimension increases. The randomized POD-rSVD-DEIM approximation follows the same qualitative behavior. The best randomized approximations are again obtained for $q = 0$, while larger values of q lead to larger nonlinear approximation errors.

The influence of the randomized parameters is summarized in Figure 11.

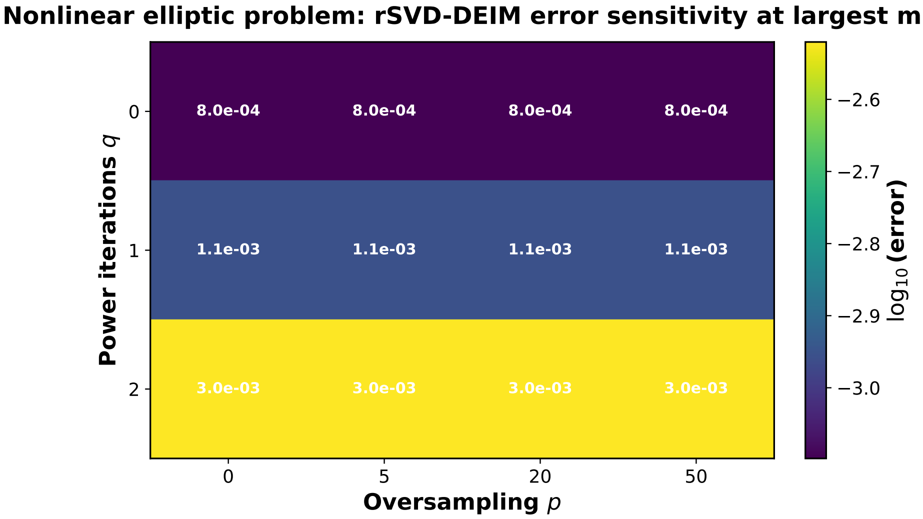

Figure 11: Sensitivity analysis of the POD-rSVD-DEIM nonlinear approximation error for the nonlinear elliptic benchmark problem. The randomized approximation remains relatively stable with respect to the oversampling parameter, while the number of power iterations has a stronger influence on the final error.

Overall, the experiments demonstrate that randomized SVD can provide highly accurate reduced bases while substantially reducing the offline basis construction cost. For both benchmark problems, the best randomized approximations are consistently obtained with $q = 0$ and moderate oversampling values. The Allen–Cahn equation exhibits an extremely favorable low-rank structure, allowing randomized methods to reproduce deterministic POD-SVD almost exactly. The nonlinear elliptic benchmark is more challenging, but randomized POD-rSVD and POD-rSVD-DEIM still provide accurate approximations with significantly reduced computational complexity.

5 Conclusions

This work investigated the theoretical foundations and practical applications of randomized Singular Value Decomposition methods. Starting from the classical theory of deterministic SVD, the study introduced randomized low-rank approximation techniques based on random projections, oversampling strategies, and power iteration schemes. The analysis focused on understanding the trade-off between computational efficiency and approximation accuracy in large-scale matrix problems.

The image compression experiments demonstrated that randomized SVD can reconstruct images with accuracy comparable to deterministic truncated SVD while significantly reducing computational cost. The numerical results confirmed that moderate oversampling values are generally sufficient to recover high-quality approximations.

The second part of the work focused on reduced-order modeling for nonlinear partial differential equations. Proper Orthogonal Decomposition, randomized POD-rSVD, POD-DEIM, and POD-rSVD-DEIM were applied to the Allen–Cahn equation and to a nonlinear steady-state elliptic benchmark problem. The experiments showed that the Allen–Cahn equation possesses a strongly compressible low-rank structure, allowing randomized methods to reproduce deterministic POD approximations almost exactly. The nonlinear elliptic benchmark problem proved to be more challenging due to the increased complexity of the nonlinear solution manifold.

A detailed sensitivity analysis was performed with respect to the oversampling parameter and the number of randomized power iterations. The experiments revealed that moderate oversampling significantly improves approximation robustness, particularly for more complex nonlinear problems. On the other hand, increasing the number of power iterations did not improve the approximation quality in the considered experiments and, in some cases, produced larger reconstruction errors. This behavior suggests that the singular spectra of the considered snapshot matrices already decay rapidly enough that additional power iterations over-amplify the dominant singular directions.

The results also demonstrated that randomized SVD and DEIM address different computational bottlenecks within reduced-order modeling. Randomized SVD substantially accelerates the offline stage associated with basis construction, while DEIM reduces the computational complexity of nonlinear evaluations during the online stage. The combined POD-rSVD-DEIM framework therefore represents a promising strategy for scalable reduced-order modeling of nonlinear systems.

Overall, the numerical experiments confirm that randomized low-rank approximation methods provide an effective balance between computational efficiency and approximation accuracy. These techniques constitute a powerful tool for modern scientific computing applications involving high-dimensional data and large-scale nonlinear dynamical systems.

Future research directions may include adaptive randomized sampling strategies, streaming and incremental POD techniques, GPU-accelerated implementations, and the application of randomized hyper-reduction methods to more complex PDE systems such as Navier–Stokes equations and multiphysics problems.

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Appendix

A Theoretical Properties of Singular Value Decomposition

This appendix presents some fundamental theoretical results concerning the Singular Value Decomposition (SVD). In particular, we discuss:

- the existence of the Singular Value Decomposition for real and complex matrices,
- the optimality of the truncated SVD approximation,
- the characterization of the approximation error in terms of singular values.

These results provide the mathematical foundation underlying both deterministic and randomized low-rank approximation methods.

A.1 Existence of the Singular Value Decomposition

We begin with the existence theorem for the Singular Value Decomposition.

Theorem 1 (Existence of the Singular Value Decomposition)**.** *Let*

$$A \in \mathbb{F}^{m\times n},$$

where

$$\mathbb{F} = \mathbb{R} \quad or \quad \mathbb{F} = \mathbb{C}.$$

Then there exist unitary matrices

$$U \in \mathbb{F}^{m\times m}, \qquad V \in \mathbb{F}^{n\times n},$$

and a diagonal matrix

$$\Sigma \in \mathbb{R}^{m\times n},$$

whose diagonal entries satisfy

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r \geq 0,$$

such that

$$A = U\Sigma V^*,$$

where V^* denotes the transpose in the real case and the conjugate transpose in the complex case.

Proof. Consider the Hermitian matrix

$$A^*A \in \mathbb{F}^{n\times n}.$$

Since A^*A is Hermitian and positive semidefinite, the spectral theorem guarantees the existence of an orthonormal basis of eigenvectors

$$v_1, \ldots, v_n$$

with corresponding eigenvalues

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0.$$

Define the singular values as

$$\sigma_i = \sqrt{\lambda_i}.$$

For every nonzero singular value σ_i, define

$$u_i = \frac{1}{\sigma_i} A v_i.$$

Then

$$\|u_i\|^2 = \frac{1}{\sigma_i^2}\langle Av_i, Av_i\rangle = \frac{1}{\sigma_i^2}\langle v_i, A^*Av_i\rangle = \frac{\lambda_i}{\sigma_i^2} = 1.$$

Furthermore, the vectors u_i are orthonormal:

$$\langle u_i, u_j\rangle = \frac{1}{\sigma_i\sigma_j}\langle Av_i, Av_j\rangle = \frac{1}{\sigma_i\sigma_j}\langle v_i, A^*Av_j\rangle = \frac{\lambda_j}{\sigma_i\sigma_j}\langle v_i, v_j\rangle.$$

Since the eigenvectors v_i are orthonormal,

$$\langle v_i, v_j\rangle = \delta_{ij},$$

thus

$$\langle u_i, u_j\rangle = \delta_{ij}.$$

Collecting the vectors into matrices

$$U = \begin{bmatrix} u_1 & \cdots & u_m \end{bmatrix}, \qquad V = \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix},$$

and defining

$$\Sigma = \operatorname{diag}(\sigma_1, \ldots, \sigma_r),$$

we obtain

$$A = U\Sigma V^*.$$

Therefore, the Singular Value Decomposition exists for every real or complex matrix. □

A.2 Optimality of the Truncated SVD

One of the most important properties of the Singular Value Decomposition is that its truncated form provides the optimal low-rank approximation of a matrix [4].

Theorem 2 (Eckart–Young–Mirsky Theorem)**.** *Let*

$$A = U\Sigma V^*$$

be the Singular Value Decomposition of a matrix

$$A \in \mathbb{F}^{m\times n},$$

with singular values ordered as

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r \geq 0.$$

Define the rank-k truncated SVD approximation

$$A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^*.$$

Then A_k is the best rank-k approximation of A in both the spectral norm and Frobenius norm:

$$\|A - A_k\|_2 = \min_{\operatorname{rank}(B)\leq k} \|A - B\|_2,$$

and

$$\|A - A_k\|_F = \min_{\operatorname{rank}(B)\leq k} \|A - B\|_F.$$

Proof. The Singular Value Decomposition can be written as

$$A = \sum_{i=1}^{r} \sigma_i u_i v_i^*.$$

The truncated approximation is obtained by retaining only the first k singular components:

$$A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^*.$$

Therefore,

$$A - A_k = \sum_{i=k+1}^{r} \sigma_i u_i v_i^*.$$

Since the singular values are ordered decreasingly, the largest singular value of the residual matrix is σ_{k+1}.

The Eckart–Young–Mirsky theorem states that no matrix of rank at most k can produce a smaller approximation error in either spectral or Frobenius norm. The proof follows from the variational characterization of singular values and the orthogonality of singular vectors.

Therefore, the truncated SVD provides the optimal low-rank approximation. □

A.3 Approximation Error of the Truncated SVD

The previous theorem implies a precise characterization of the approximation error.

Theorem 3 (Truncated SVD Approximation Error)**.** *Let*

$$A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^*$$

denote the rank-k truncated SVD approximation of A.

Then:

$$\|A - A_k\|_2 = \sigma_{k+1},$$

and

$$\|A - A_k\|_F = \left(\sum_{i=k+1}^{r} \sigma_i^2 \right)^{1/2}.$$

Proof. From the SVD expansion,

$$A - A_k = \sum_{i=k+1}^{r} \sigma_i u_i v_i^*.$$

The singular values of the residual matrix are therefore

$$\sigma_{k+1}, \sigma_{k+2}, \dots, \sigma_r.$$

Since the spectral norm equals the largest singular value,

$$\|A - A_k\|_2 = \sigma_{k+1}.$$

Similarly, the Frobenius norm satisfies

$$\|A - A_k\|_F^2 = \sum_{i=k+1}^{r} \sigma_i^2,$$

which implies

$$\|A - A_k\|_F = \left(\sum_{i=k+1}^{r} \sigma_i^2 \right)^{1/2}.$$

This concludes the proof. □

B Theoretical Foundations of Randomized Singular Value Decomposition

This appendix presents the main theoretical ideas underlying Randomized Singular Value Decomposition (rSVD). The goal is not to reproduce the complete probabilistic analysis developed in [3], but rather to provide the mathematical intuition and the principal theoretical results that justify randomized low-rank approximation methods.

The randomized framework is based on the observation that a low-rank matrix can be accurately approximated by projecting it onto a randomly generated low-dimensional subspace. These methods combine random sampling, orthogonal projection, and classical deterministic factorizations to obtain efficient approximations of the dominant singular subspaces.

The presentation in this appendix follows the framework introduced by Halko, Martinsson, and Tropp in [3].

B.1 Low-Rank Approximation Framework

Let

$$A \in \mathbb{R}^{m \times n}$$

be a matrix whose singular values decay sufficiently rapidly.

The central idea of randomized low-rank approximation is to construct a matrix

$$Q \in \mathbb{R}^{m \times \ell},$$

with orthonormal columns, such that

$$A \approx QQ^T A. \tag{73}$$

The matrix Q approximates the dominant column space of A. Once this low-dimensional subspace has been identified, the matrix can be compressed into a much smaller representation:

$$B = Q^T A. \tag{74}$$

A deterministic factorization is then computed on the reduced matrix B, yielding an approximate decomposition of the original matrix.

This two-stage structure is one of the key ideas introduced in [3].

B.2 Random Sampling and Range Approximation

The randomized approach constructs the basis Q using random projections.

A random test matrix

$$\Omega \in \mathbb{R}^{n \times (k+p)} \tag{75}$$

is generated, typically with independent Gaussian entries:

$$\Omega_{ij} \sim \mathcal{N}(0, 1). \tag{76}$$

The sampled matrix is then computed as

$$Y = A\Omega. \tag{77}$$

The columns of Y consist of random linear combinations of the columns of A. With high probability, these samples span the dominant singular subspace associated with the largest singular values of A.

An orthonormal basis is then extracted using QR factorization:

$$Y = QR. \tag{78}$$

The matrix Q approximates the range of A, allowing the projection

$$A \approx QQ^T A. \tag{79}$$

The fundamental intuition is that random vectors are unlikely to miss dominant directions in high-dimensional spaces. Consequently, random projections preserve most of the relevant information contained in matrices with rapidly decaying singular values.

B.3 Oversampling Parameter

The parameter

$$p \geq 0$$

is called the oversampling parameter. Instead of computing exactly k random samples, the algorithm computes $k + p$ samples.

The purpose of oversampling is to improve the probability that the sampled subspace captures the dominant singular directions of the matrix. In practice, small values such as

$$p = 5 \quad \text{or} \quad p = 10$$

are often sufficient to obtain highly accurate approximations.

Oversampling becomes particularly important when:

- the singular values decay slowly,
- the matrix is noisy,
- the singular spectrum contains clustered values,
- the target rank is small.

Increasing p enlarges the sampled subspace and therefore improves robustness, although at the cost of additional computations.

B.4 Power Iteration Scheme

When the singular values decay slowly, the basic randomized projection may not sufficiently separate dominant and non-dominant singular directions. To improve the approximation quality, randomized methods often employ a power iteration scheme.

Instead of directly computing

$$Y = A\Omega,$$

the algorithm computes

$$Y = (AA^T)^q A\Omega, \tag{80}$$

where

$$q \geq 0$$

is the number of power iterations.

Using the Singular Value Decomposition

$$A = U\Sigma V^T,$$

it follows that

$$(AA^T)^q A = U\Sigma^{2q+1} V^T. \tag{81}$$

Therefore, the singular values become

$$\sigma_i^{2q+1}.$$

The power iteration amplifies the separation between dominant and smaller singular values. Consequently, the approximation becomes more accurate for matrices with slowly decaying spectra.

However, increasing q:

- increases computational cost,
- requires additional passes over the matrix,
- may amplify numerical round-off errors.

For many practical applications, values

$$q = 1 \quad \text{or} \quad q = 2$$

already provide substantial improvements.

B.5 Probabilistic Error Bounds

One of the most important theoretical results in randomized low-rank approximation concerns the approximation error of the projected matrix.

Let Q denote the orthonormal basis obtained through randomized sampling. Then the approximation error satisfies

$$\|A - QQ^T A\|_2 \approx \sigma_{k+1}, \tag{82}$$

which is the optimal error associated with the truncated SVD.

Halko, Martinsson, and Tropp proved that randomized algorithms achieve near-optimal approximations with high probability.

A simplified version of their result states that

$$\mathbb{E}\left[\|A - QQ^T A\|_2\right] \leq \left(1 + \frac{4\sqrt{k+p}}{p-1}\sqrt{\min(m,n)}\right)\sigma_{k+1}. \tag{83}$$

This result implies that the randomized approximation error differs from the optimal truncated SVD error only by a moderate multiplicative factor.

Furthermore, the probability of large deviations decreases extremely rapidly with increasing oversampling parameter p. As discussed in [3], the failure probability decreases superexponentially with respect to p.

B.6 Computational Complexity

One of the principal advantages of randomized methods is the reduction of computational complexity.

For a dense matrix

$$A \in \mathbb{R}^{m \times n},$$

classical truncated SVD typically requires

$$\mathcal{O}(mnk) \tag{84}$$

floating-point operations.

In contrast, randomized SVD approximately requires

$$\mathcal{O}(mn\log(k)) \tag{85}$$

operations in many practical settings.

Randomized algorithms are particularly advantageous when:

- the matrix is very large,
- only a small number of singular components is required,
- the matrix is sparse,

- the data are streamed or stored out-of-core,
- parallel architectures are available.

Moreover, randomized methods require only a small number of passes over the data matrix, making them particularly attractive for modern large-scale scientific computing applications.

B.7 Discussion

The theoretical results presented in this appendix explain why randomized low-rank approximation methods have become increasingly important in modern numerical linear algebra.

The success of randomized SVD relies on several key observations:

- many practical datasets possess low-dimensional dominant subspaces,
- random projections preserve most of the relevant geometric information,
- oversampling substantially improves robustness,
- power iterations improve spectral separation,
- the approximation error remains close to the optimal truncated SVD error with high probability.

These properties allow randomized algorithms to combine strong theoretical guarantees with excellent practical performance.

In the numerical experiments presented throughout this work, the effectiveness of randomized SVD was confirmed in both image compression and reduced-order modeling applications, where randomized methods successfully reproduced deterministic low-rank approximations while significantly reducing computational cost.